The Korean Society for Language and Information
http://dx.doi.org/10.29403/LI.22.2.1

Predictive Representation for Upcoming Linguistic Input in Younger and Older Adults*

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Hongoak Yun, Sun Hyun Moon and Soo Rim Noh. 2018. Predictive Representation for Upcoming Linguistic Input in Younger and Older Adults. *Language and Information*, 22.2, 1-25. The immediate use of the contextual information to predict the upcoming information during sentence comprehension may vary depending on the aspects of the linguistic and world knowledge and the cognitive ability of readers. In this study, we investigated how well older readers employ their linguistic and world knowledge to develop the expectation for the upcoming information during sentence comprehension. In two offline tasks (cloze and listing), older and young readers were asked to complete sentence fragments (when agents and recipients were given or when agents and patients were given). The responses were used to compute predictability measurements (structural and lexical predictability). We found no aging effect in the production of likely structural choices and likely lexical choices, meaning that the most predictable ones to older readers were also the most predictable to young readers. Significant aging effect emerged in the production of unlikely choices; older readers were more predictable to unlikely structures but less predictable to unlikely words. We discuss the contextual sensitivity of older readers in formulating their knowledge representation for what is coming up next.

Key words: predictability, aging, knowledge representation, contextual sensitivity

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* This work was supported by the Ministry of Education of the Republic of Korea and the National Research of Korea(NRF-2017S1A3A2066319)

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1. Introduction

Predictability in sentence comprehension has been one of the key notions in recent psycholinguistic theories. It suggests that while reading a sentence, readers automatically develop their expectation for what is coming up next and apply their expectation immediately and actively in the processing of the upcoming word, even before they actually encounter it. For example, given the sentence fragment (1), readers are most likely to expect to encounter *groom* for the upcoming position, probably due to the use of world knowledge, although other possible choices (e.g., *friend, baby,* or *stick*) are semantically and pragmatically plausible for the position. Thus, when the phrase, *groom,* actually appears, it becomes extremely easy to integrate the most probable phrase into the sentence fragment.

(1) The bride put on a wedding ring on the ___________

In the language processing literature, the issues of predictability effect have been explored in an enormously large number of studies. As referred in Huettig (2015), the question of WHAT, what (i.e., which word or phrase) is predicted or what (e.g., contextual information) leads to prediction, has been examined. For example, studies in self-paced reading comprehension reported that predictable words of higher cloze probability are read faster than less predictable words of lower cloze probability (Roland et al. 2012; Yun and Hong 2014). Eye-movement reading studies observed that predictable words have decreased fixation durations and fewer regressive eye movements (Ashby et al. 2005; Balota et al., 1985; Ehrlich and Rayner 1981; Frisson et al. 2005; Frisson et al.,. 2017; Kliegl et al., 2004; Rayner et al., 2006; Staub 2011; Steen-Baker, et al., 2017; Veldrey and Andrews, 2018). The similar results were also demonstrated in the studies of neurological methods like event-related potentials (ERPs) where the N400 was smaller for the more expected, given the prior context, relative to the less expected (Bicknell et al. 2010; DeLong et al. 2014; DeLong et al. 2005; Federmeier et al. 2007; Kutas and Federmeier, 2011; Kwon et al., 2017; Otten and Van Berkum 2008; Wlotko et al., 2012; Van Berkum et al. 2005; Van Petten and Luka 2012). The question of WHEN, when predictability effect emerges, has long been debated under the notion of anticipation versus integration (DeLong et al., 2014; Kamide, 2008; Kwon et al., 2017; Lau et al., 2006; Wicha et al., 2003; and many more eye-tracking studies using visual world paradigm including Altman and Kamide, 1999; Kamide et al., 2003; Knoeferle et al., 2004; Tanenhaus et al., 1995). However, in comparison to the questions of WHAT and WHEN, the question of HOW, the mechanism of how predictability effect is elicited, has not received much attention. This paper touches the issues of HOW with a special focus of individual differences in the predictability effect, by comparing older readers to young readers.

The issue of HOW covers questions like how or under which systems the predictability effect is implemented through readers' mental processes. Suppose Example (1) above, the predictive processing in (1) requires readers to go through several systems: First, while reading the given context (i.e., *The bride put on a wedding ring on the* ...), readers automatically pre-activate linguistic input associated with *groom* such as its grammatical category (e.g., noun), semantic features (e.g., being elongated, round, or small), and pragmatic knowledge (e.g., wedding event). Namely, they should be able to exploit related linguistic and world knowledge, cued from the given context, and employ their knowledge automatically to generate their expectation for the yet-to-encounter information. Second, in addition to their competence in establishing readers' knowledge representation of uncertainty, readers also need to have good-enough cognitive resources like working memory or cognitive control in applying their expectation in the processing of an upcoming word while they are busy in integrating all other relevant information into sentences. To put it brief, the mechanism of predictive sentence comprehension (i.e., how it works) is not simple.

Several proposals have attempted to account for the complexity of predictability effect in comprehension in the way that multiple systems of mental processes and their interactions need to be carefully taken into account. Some researchers proposed a two-system account (Kuperberg, 2007) with an idea that the human mind makes use of two distinct systems: 1) a quick and automatic system which is sensitive to detect semantic associative relationships between incoming words and information stored within semantic memory (knowledge) and 2) a strategic system for combinatorial stream which is sensitive to multiple linguistic constraints and higher-order meaning in the integration of an upcoming word into a sentence. This proposal has been in line with recent cognitive theories (c.f., Kahneman, 2011; Stanovich and West, 2000) like System 1 which operates automatically and quickly with little effort and no sense of voluntary control and System 2 which allocates attention to effortful mental activities that demand it, including complex computations.

The neutral evidence supporting for the existence of the quick and automatic system has been found in semantic priming studies which indicate temporal and inferior prefrontal regions are sensitive to associative relationships (e.g. Copland et al., 2003; Kotz et al., 2002; Matsumoto et al., 2005; Rossell et al., 2003), evoking the N400 in ERP studies (Hagoort et al., 2004; Kiehl et al., 2002; Kuperberg et al., 2003). System 2 is thought to involve posterior inferior frontal cortices, motor and parietal cortices, and middle and superior prefrontal cortices (Friederici et al., 2003; Kuperberg, 2007; Kuperberg et al., 2003; Ni et al., 2000). These regions are considered to be accompanied by active control and attention (Duncan, 2010) and they seem to be activated by both morpho-syntactic violations and semantic-thematic violations, evoking the P600s (Kuperberg et al., 2003; Kuperberg, 2007).

An important claim is that despite the partial distinction between System 1 and System 2, these systems are thought to constantly interact. Kuperberg (2007) for instance argues that the balance between the two systems is influenced by task demands, context, and individual differences such as working memory. Huettig (2015) claimed that factors like working memory or processing speed plays a role as a mediator in predictive language processing. Crucially, by manipulating gender-marked information on articles in Dutch with the hypothesis that readers could use article gender information to predict the upcoming word, Huettig and Janse (2013) observed that anticipatory eye movements (i.e., eye fixations landed to visual targets prior to their corresponding auditory inputs) were modulated by readers' working memory abilities and processing speed. The researchers claimed that working memory and processing speed play a role in sentence processing in that they index cognitive proficiency with which can retrieve information from long-term memory and with which integrate unfolding information into the representation of the sentence meaning.

Despite of its importance, the influence of such mediating factors on prediction in language processing has not been systematically investigated. We strongly agree with Huettig and Janse (2016) in their saying that "theoretical models of predictive language processing will only be complete if they can fully account for the interplay of individual differences in the mediating factors and the mechanisms of prediction (pp. 81)." However, only few studies have attended on this unsolved issue.

We suggest that the success of the issue (i.e., the role of mediator in multiple systems for predictive sentence comprehension) relies on any evidence hinting the dissociations (or discreteness) of multiple systems. In other words, mental processes which contribute to come up with upcoming linguistic representation extracted from previous context does not employ the exact same process which attributes to the rapid use of automatically-developed expectation. For example, the predictive effect in comprehension is relatively weaker to L2 readers compared to L1 readers (Dussias et al., 2013; Grüter et al., 2012; Grüter and Rohde, 2013; Hopp, 2013), partially due to L2 readers' poor linguistic proficiency enough to develop linguistic expectation. However, the weak predictive effect observed in the L2 studies should not be reasoned similarly to the weak predictive effect that older readers showed in predictive language processing. Presumably, the difficulty that older readers are faced with is mostly due to their cognitive decline, rather than their linguistic knowledge (Wlotko et al., 2010). That being said, the two groups, L2 readers and older readers, look to have similar difficulty (i.e., weak predictive effect) but their difficulties may underlie different reasons.

Our ultimate goal is to observe evidence for the discreteness of multiple systems of mental processes in predictive sentence comprehension. In particular, we aim to examine the cases in which one system is intact but the other system is damaged, by taking older readers as our research target groups. In our first step, we investigate whether older readers would develop upcoming information, as good as young readers, by employing their linguistic and world knowledge.

2. Aging and Language Comprehension

Many changes occur in various aspects of perception, memory, and learning along with changes in the body and sensory organs due to normal aging (Schaie et al., 1996). Typically, both growth and declines are involved at all points in human development, and it is the same for cognition (Baltes et al., 1999). The patterns of cognitive aging can be summarized as follows. While fluid intelligence that includes speed, working memory, and long-term memory shows gradual age-related declines beginning in younger adulthood, crystallized intelligence that includes verbal knowledge, wisdom, and practical problem solving is relatively well preserved even in old age because they are accumulated through cultural and educational experiences throughout the life course (Baltes et al., 1999; Park et al., 2002; Schaie, et al., 1996).

These different trajectories of cognitive development bring out remarkable changes in language processing. The reading speed of older adults becomes slower than that of younger adults due to the reduction in overall processing speed with age, but lexical processing or word recognition has been found to be relatively well preserved in old age (Payne et al., 2012; 2014). The level of comprehension of older readers is lower than younger readers for sentences with complex semantic processing that increase the load of working memory, and even though deterioration is also observed in the amount of recall after reading sentences, it was found that older readers rely more on knowledge to form a higher level gist representation (Stine-Morrow et al., 2006). Thus, it appears that crystallized intelligence and informal language experience that have been accumulated throughout the lifespan increase the efficiency of word and syntactic processing. Moreover, changing reading strategy to knowledge-driven processing in late adulthood has been suggested to compensate the decline in reading processing due to declining fluid intelligence (Stine-Morrow et al., 2008).

Domestic studies that have investigated the sentence processing of the elderly also have found that the speed of older readers is slower than that of younger readers, and older readers were more inaccurate and slower compared to younger readers when the noun-phrase type of main clause in a relative clause sentence was inconsistent (e.g., Kim and Lee, 2015; Lee and Sung, 2015). There were, however, no age-related differences in using mental simulation during sentence comprehension (i.e., verifying pictures that matched the shape implied by the target sentence) in the study by Noh et al., (2017). That is, similar to the findings of studies conducted in the western countries, domestic studies also found that sentence processing ability relying on knowledge was preserved even though reading speed and the ability of processing complex sentences that relies on working memory were found to be decreased due to cognitive decline with age.

That being granted, how will the predictability effects appear in processing words during sentence reading according to normal aging? The findings from the studies using behavioral paradigms have revealed that older adults facilitated the recognition and integration of words by discriminatingly using contextual information as good as younger adults (e.g., Stine and Wingfield, 1994; Lash, et al. 2013; Madden, 1988; Stine-Morrow et al., 2008). In the ERP studies, however, the use of predicted words through real-time context by older adults were found to be neither active nor immediate since they showed reduced N400 effects of sentence context compared to younger adults (e.g., Wlotko et al., 2010). Similarly, a recent eye tracking study has reported that age-related increase in context sensitivity occurs after initial fixation on target word (i.e., later stage of processing). Steen-Baker et al. (2017) reported comparable effects of contextual constraints for younger and older adults in early word-recognition processes, as measured by first fixation durations and the probability of skipping, whereas the effect of context on later-integrative processes was greater in older rather than in younger adults. In other words, older readers when compared to their younger counterparts, spent differentially more time in rereading (as indexed by longer regression path durations) when encountering unexpected rather than expected sentence-final target words. Steen-Baker et al. (2017) argued that while the ability of using sentence context for lexical processing remains intact with advancing age, context sensitivity for meaning integration increases with age. Thus, studies using time-sensitive measures such as ERPs and eye tracking show that the strategy of using context in reading changes according to aging. A further support from fMRI study by Peelle et al., (2010) that assuming that good connectivity between different neural processing regions may contribute to a high speed of information integration, neural connectivity in older adults was reduced compared to younger adults.

There are two important aspects about older readers: First, their fluid intelligence (e.g., processing speed, working memory, and long-term memory) is gradually declined, whereas their crystallized intelligence (e.g., verbal knowledge, wisdom, and practical problem solving) are relatively well preserved. Second, older readers' abilities of using sentence context for lexical processing and context sensitivity for meaning integration are intact but these abilities are used during online processing in particular reading strategies. We also note the speculation of Hutteg and Janse (2013) that mediating factors in multiple systems might interact in particular ways with different mechanism of prediction. Perhaps working memory capacity is more important for combinatorial mechanisms of prediction (cf. Kuperberg, 2007) than for simple associative mechanisms (cf. Kuperberg, 2007; Pickering and Garrod, 2013). A tentative conclusion, then, is that the difficulty observed in older readers' predictive sentence processing is fundamentally dependent on their declined working memory, which in turn leads to more difficulty of operating the combinatorial system (System 2) but less difficulty of operating the fast and automatic system (System 1). Therefore, if so, it is speculated that the knowledge and linguistic competence of older readers to establish linguistic representation for upcoming information would be as good as those of young readers. In the present study, as mentioned above, we begin our investigation by examining the issues related to System 1, that is, whether older readers would develop upcoming information, as good as young readers, by employing their linguistic and world knowledge.

3. The Present Study

To test our hypothesis, we conducted two paper-and-pencil studies which aimed to examine readers' abilities to build up the linguistic representation for upcoming information by using Korean dative sentence fragments that Yun and Hong (2014) has used, as shown in (2a-b). As we hypothesized, if older readers' knowledge to make use of the linguistic and contextual information is well preserved, in other words, if older readers' fast and automatic system to detect semantic properties for upcoming information by exploiting relevant information from the context is intact. we expect to observe that older readers' predictive representation for the upcoming information would not be significantly different from young readers' predictive representation for it.

(2a) Chelswu-ka kyengchal-eykey ____________
Chelswu-$_{\text{NOM}}$ policeman-$_{\text{DAT}}$
(2b) Chelswu-ka sinpwunchun-ul ____________
Chelswu-$_{\text{NOM}}$ ID card-$_{\text{ACC}}$

From the data obtained from young readers, Yun and Hong (2014) demonstrated that it was highly likely to encounter a theme/patient role after the sentence fragments like (2a) in which an agent and a recipient were introduced, whereas it was to see a verb after the sentence fragments like (2b) in which an agent and a theme/patient were introduced. Crucially, in spite of no grammatical violation, readers' expectation to see a recipient role coming up after the fragments like (2b) was extremely low. Together with the predictability about which structure is likely to appear next, so-called structural predictability (role predictability), the researchers also estimated which exact word corresponding to the upcoming role position would like to appear for a given context, so-called lexical predictability (role filler predictability). By controlling for the degree of structural and contextual predictability, Yun and Hong (2014) observed behavioral evidence supporting for the validity that the predictive representation generated by young readers well accounted for readers' behaviors. Furthermore, the Yun and Hong's results were also replicated in the eye movement reading comprehension conducted by Yun et al. (2017a).

In the present study, we conducted two offline tasks (cloze task, listing task) to older readers and we estimated two predictabilities (structural predictability, lexical predictability) using two types of computation measurements (cloze probability, uncertainty). Then, we compared all estimated predictability measurements computed based on offline responses obtained from two age groups (older readers, young readers).

4. Participants

For young adults, we used the same data that Yun and Hong (2014) collected. 40 undergraduate students participated in these tasks and all were normal with no difficulty in vision and language processing. The data were reanalyzed for the purpose of the present study and were compared to the data from older participants.

Twenty-nine older participants were recruited through advertisements in the community and welfare centers located in the City of D. All participants spoke Korean as their mother tongue. The participants who had the following conditions were excluded: people who underwent treatment or operation after being diagnosed with brain damage or brain disorder; people who were diagnosed with a psychiatric illness such as depression and dementia; and people who underwent treatments through medication or psychiatric counseling due to depression within a recent year. General demographic information of the participants is presented in Table 1. The average age of the participants is 68 years of age (range = 56-80, SD = 6.38). The gender of the participants is comprised of 41.4% men and 58.6% women. As for education level, 55.2% of the participants had a high school diploma and 27.5% of people had a college degree or higher education.

Table 1. Characteristics of the Participants

Variable		Number	%
Age	Range	56~80	
	M(SD)	68.34(6.38)	
Sex	Male	12	41.4
	Female	17	58.6
Education	Elementary school	2	6.9
	Middle school diploma	3	10.3
	High school diploma	16	55.2
	Bachelor's degree	8	27.5
K-MMSE	*M(SD)*	27.31(1.39)	
K-WAIS-IV vocabulary	*M(SD)*	27.55(6.88)	

Notes. K-MMSE=Korean Mini-Mental State Examination. K-WAIS-IV=Korean Wechsler Adult Intelligence Scale.

To evaluate the overall cognitive function of the older participants, the Korean Mini-Mental State Examination (K-MMSE; Kwon & Park, 1989) was administered. K-MMSE consists of 19 questions that evaluate the level of orientation, verbal memory registration, naming, visuospatial skills, and comprehension. The full score on this test is 30 points, and the scores above 25 are classified as no cognitive impairment. All the older participants scored 25 points and higher, with the average score of 27.31 (SD = 1.39), showing no cognitive impairment. In addition, older participants' verbal ability was evaluated using the vocabulary test from the Korean Wechsler Adult Intelligence Scale (K-WAIS)-IV. On this test, a list of words is shown to the participants, and their word definition is scored from 0 to 2 points based on accuracy and concreteness. The full score on the vocabulary test is 57 points; according to Kwak (2016)'s study, the average score of people with over 12-year education was 31.9 (SD = 8.34). In the present study, the average score of the older participants was 27.55 (9-41, SD = 6.88).

5. Stimuli and Procedure

Cloze Task. A cloze task was conducted in order to estimate the degree of structural predictability, by using the sentences like (3a-b) that Yun and Hong (2014). Recipient NPs followed by agent NPs were presented in sentence fragments like (3a), whereas theme NPs after agent NPs were presented in sentence fragments like (3b). The participants completed sentence fragments by filling in the cloze on their own, following the Korean grammar rules. In addition to 61 target sentences, 30 filler sentences were included to prevent participants from being strategic in their responses. Thus, the structures for filler sentences were diverse such as subordinate clauses, adverbial clauses, and relative clauses.

(3a) Chelswu-ka kyengchal-eykey ____________
Chelswu-$_{\text{NOM}}$ policeman-$_{\text{DAT}}$
(3b) Chelswu-ka sinpwunchun-ul ____________
Chelswu-$_{\text{NOM}}$ ID-$_{\text{ACC}}$

Listing Task. A listing study was conducted to estimate the degree of lexical predictability pertinent to the upcoming word for a given event. For this listing task, the dative sentences used in the cloze task were used. Unlike the sentence fragment used in the cloze task, participants were explicitly asked to produce words associated with target thematic roles. To be more specific, for sentence fragments like (4a), participants had to come up with three specific patients/themes role fillers that were suitable for the given context. On the other hand, for sentence fragments like (4b), participants had to list three specific recipients that were relevant for the given contexts. No filler sentences were included.

(4a) Chelwu-ka kyengchal-eykey	________ ul/lul
	________ ul/lul
	________ ul/lul
Chelswu-$_{\text{NOM}}$ policeman-$_{\text{DAT}}$	________ $_{\text{ACC}}$
(4b) Chelwu-ka sinpwunchun-ul	________ eykey
	________ eykey
	________ eykey
Chelswu-$_{\text{NOM}}$ ID card-$_{\text{ACC}}$	________ $_{\text{DAT}}$

The experiment was conducted in the following: a survey of participants' demographic information, K-MMSE, a cloze task, a listing task, and vocabulary test. For the cloze and listing tasks, the participants were asked to complete the sentences freely. Prior to performing the listing task, the participants were given the instruction on the experiment method, the same for the previous task, with an additional caution to avoid using pronouns and proper nouns indicating specific names of people and brands. During the experiment, the researchers observed the participants' progress to check their level of understanding of the sentences and provided assistance when necessary.

6. Analysis and Results

Analysis. To measure the goodness of readers' linguistic and contextual knowledge for the upcoming information, we employed two kinds of predictability measurements (i.e., cloze probability and uncertainty) in two types of predictability (i.e., structural predictability and lexical predictability) in the way that Yun et al. (2017b) demonstrated.

Equation (1), negative log-transformed cloze probability, represents the claim, proposed by the Surprisal model (Hale, 2001; Levy, 2008), that the degree of difficulty in the integration of a word into a sentence is proportional to the degree to which the word is predicted for a given context. For example, in sentences like *The horse raced past the barn fell*, readers become enormously slowed down when they encounter the unexpected main verb of extremely low conditional probability, *fell,* at the sentence-final position (Hale, 2001). Thus, computing the degree of surprisal associated with a word is one way to estimate the goodness of readers' predictability in comprehension. In the present paper, we simply take the part of conditional probability without taking log-transformation, as shown in Equation (2).

Equation (1) $difficulty \propto -\log P(w_i | w_{i...i-1}, Context)$

Equation (2) $P(w_i | w_{i...i-1}, Context)$

Equation (3), known as Shannon's Entropy, indicates the extent of uncertainty based on the probability distribution of possible choices that could occur at time *t* at a sentence. This probabilistic measurement indicates that the difficulty in the integration of a word into a sentence is affected by the extent of uncertainty that exists at the position that the word appears. The key notion of uncertainty is that it is important to take into account the probability of a target word and those of other possible words that could occur instead. If a context is certain (i.e., low uncertainty), this means that there is only a few possible choices that are highly likely to occur. That is, the context of low uncertainty is predictable. In contrast, if a context is uncertain (i.e., high uncertainty), that means there are a number of possible choices that can appear, suggesting that the context of high uncertainty is less predictable.

Equation (3) $H(X; w_1^t) = -\sum_{x \in X} P(x | w_1^t) \log P(x | w_1^t)$

We used both types of predictability measurements: (1) Cloze probability, expressed in Equation (2), indicates how likely a specific choice is to encounter out of all the other possible choices; (2) Uncertainty, expressed in Equation (3), indicates how likely other possible choices would appear for a given context, suggesting how strongly or weakly the contextual information is constraining the range of possible choices. At first, we coded the responses obtained from old and young readers by their structural information (cloze task) and by lexical information (listing task). Then, we computed the cloze probability of a target and the uncertainty associated with the target. Table 2 displays the layout of our analyses and computations.

For our structural coding of the cloze data, participants' responses were marked by their grammatical category and thematic roles. For example, if *sinpwunchun-ul* (ID card) was produced for the fragment, *Chelwu-ka kyengchal-eykey* (Chelwu-$_{NOM}$ policemen-$_{DAT}$ ...), *sinpwunchun-ul* was coded as Theme NP. In the lexical coding of the listing data, participants' responses were coded as they are, but some of the lexical items were coded as the same (e.g., both *emoni*(mother) and *emma*(mom) were coded as *emoni*).

Table 2. The layout of our predictability measurements

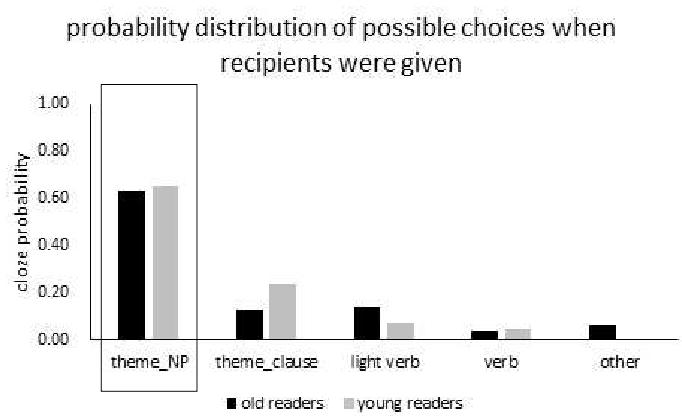
Task	Task	Predictability measurements	Group comparison
cloze	structural predictability	cloze probability	Old vs. Young readers
		uncertainty	Old vs. Young readers
listing	lexical (contextual) predictability	cloze probability	Old vs. Young readers
		uncertainty	Old vs. Young readers

Results. We first report the results of structural predictability and then move on to reporting those of lexical predictability.

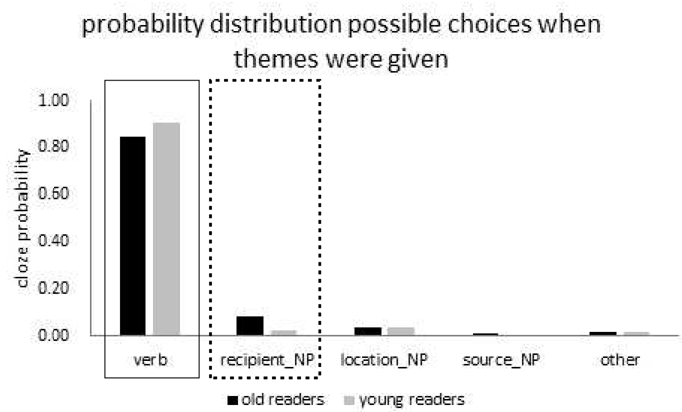
First, the structural probability and uncertainty were estimated based on the completions of the cloze task (see Figure 1a-b). When recipients were presented (Figure 1a), theme NPs were produced most, indicating that readers highly expect to encounter theme NPs. However, when patient NPs (Figure 1b) were presented, recipient NPs were not produced much but verbs were produced most, meaning that verbs are mostly likely to be encountered. Crucially, as we hypothesized, these prediction patterns would be similar between the both age groups. Older readers, like young readers, generated theme NPs most after the combinatorial information of agents and recipients and did verbs most after the combinatorial information of agents and themes. The results from independent t-tests were illustrated in Table 3.

When recipients were presented, the degree of target probability (theme NPs), estimated using Equation (2), were not significantly different across age groups (t(70) = -1.66, p = .88), meaning that both age groups strongly expected to encounter theme NPs as the upcoming structural choice. However, the degree of structural uncertainty, estimated using Equation (3) was higher among the older group ($t(70) = 3.88$, $p < .001$), suggesting that the probability distribution of possible structural choices generated from older readers were less certain compared to that generated from young readers. Although the differences of structural uncertainty were statistically significant across the age groups, we hesitated to take the result as being statistically-wise relevant in that the differences were mostly like to emerge due to the distribution of unlikely choices with the target (theme NP) being predominantly predictable in both age groups.

(a)



(b)



[Figure 1a-b] The probability distribution of possible structural choices for the upcoming position (a) when NPs associated with recipients were given and (b) when NPs associated with themes/patients were given. A solid-lined box indicates target structural choice (Figure 1a-b). A dotted-line box indicates the most-likely structural choice (Figure 1b).

When themes were presented, the predictability of most-likely choice (verb) were extremely high in both groups with no aging effect ($t(45) = -1.72$, $p = .09$). Similarly, the degree of structural uncertainty was not significantly different between age groups ($t(45) = 1.44$, $p = .16$). Target probability (recipient NPs) were extremely low in both groups but there observed significant aging effect ($t(45) = 2.51$, $p < .05$), meaning that the predictability of targets were higher to older readers than young readers. Again, despite the statistical differences of target probability, we are careful to say whether the probability difference would lead to actual difference in online reading.

Table 3. Comparisons of predictability measurements in two sentence fragments across age groups

Structure	Predictability	Age	Results	
Recipient being presented	Structural Uncertainty	old	$M = .45$, $SD = .11$	$t(70) = 3.88$, $p < .001$
		young	$M = .36$, $SD = .08$	
	Target Probability (theme NPs)	old	$M = .64$, $SD = .12$	$t(70) = -1.66$, $p = .88$
		young	$M = .64$, $SD = .15$	
Patient being presented	Structural Uncertainty	old	$M = .20$, $SD = .12$	$t(45) = 1.44$, $p = .16$
		young	$M = .15$, $SD = .13$	
	Target Probability (recipient NPs)	old	$M = .08$, $SD = .09$	$t(45) = 2.51$, $p < .05$
		young	$M = .03$, $SD = .04$	
	Probability of most-likely choice (verb)	old	$M = 1.12$, $SD = .11$	$t(45) = -1.72$, $p = .09$
		young	$M = 3.91$, $SD = .46$	

Second, we found the commonality and difference in the lexical (contextual) predictability between the age groups. We observed the patterns of the most-likely lexical choices for given contexts were fairly similar between older readers and young readers. When recipients were given, the most predictable themes (i.e., Number 1 choice) that older readers generated were the exactly same with those that young readers generated in 21 contexts out of 36 contexts (58%). The overlapping percentages increased when we extended our comparisons into the range of Number 2 or 3 choices at about 75%. The similar patterns were also found when themes were provided, the most predictable recipients that older readers generated were the exactly same with those that young readers generated in 12 contexts out of 24 contexts (50%) and the percentages went up when the comparisons were widely extended (80%). Simply saying, these results suggested that the most primed lexical choices by contexts were not different across the age groups: For given contexts, the most-likely words to older readers were the most likely ones to young readers, too.

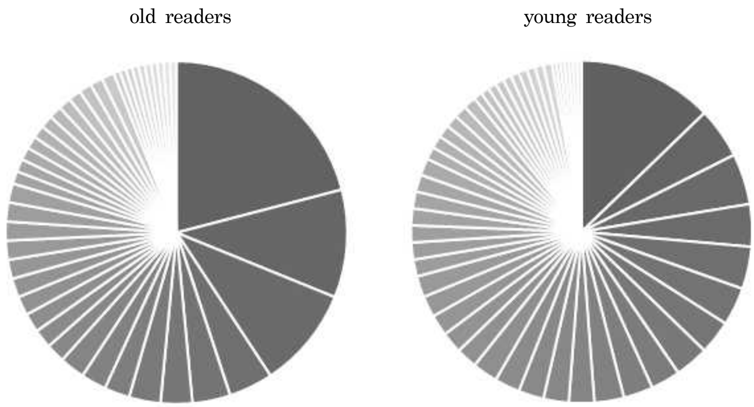
We also observed the differences of generated lexical choices between the age groups, in particular, in terms of the level of lexical uncertainty. Figure 2a illustrates the probability distribution of possible words for the patient position when a recipient was given (e.g., *Chelswu-ka kyengchal-eykey* meaning Chelswu-$_{\mathrm{NOM}}$ policeman-$_{\mathrm{DAT}}$). Figure 2b displays the possible distribution of possible words for the recipient position when a theme was given (e.g., *Chelswu-ka sinpwunchun-ul* meaning Chelswu-$_{\mathrm{NOM}}$ ID-$_{\mathrm{ACC}}$). Note that the distribution of older readers consists of a couple of highly-likely lexical choices and less number of possible lexical choices overall, compared to that of young readers. The results of independent t-test revealed that the lexical uncertainty generated by older readers were significantly lower than that by young readers: When recipients were given, as in Figure 2a, the lexical uncertainty of older readers (M = 1.49, SD = .12) were lower than that of young readers (M = 4.96, SD = .44). The difference of the level of lexical uncertainty was significant between the groups ($t(70) = 45.96$, $p < .001$). Similarly, when themes were given, as in Figure 2b, the lexical uncertainty of older readers (M = 1.11, SD = .11) were lower than that of young readers (M = 3.91, SD = .46). The difference of lexical uncertainty was significant between the groups ($t(46) = 28.81$, $p < .001$).

These results suggest that the same context were considered to be more predictable by older readers than young readers. Presumably, older readers tended to make previous contexts more constraining than young readers,[1] which in turn probably leads to the

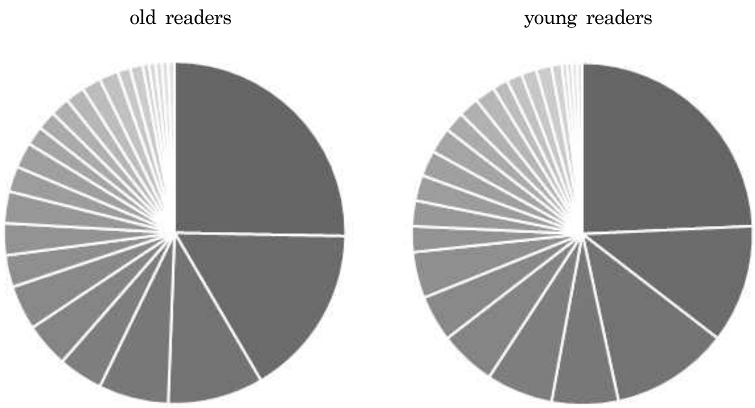
1 One of the reviewers asked that the lower lexical uncertainty may appear because older readers fail to activate possible words in efficiency due to their lower cognitive competence. In fact, it is not clear whether the results refer to older readers' constraining use of the contextual information, whether they show older readers' low cognitive competence in leading to poor activation of possible word choices, or whether

prediction that older readers might be efficiently sensitive in using the contextual information in the processing of the upcoming word during online processing.

(a) An example of lexical uncertainty when recipients were presented



(b) An example of lexical uncertainty when themes were presented



the constraining use of the contextual information may represent a typical strategy for readers of poor cognitive competence to deal with the context. However, in either way, our results clearly means lexical representation generated by old readers are more constraining than the one generated by young readers.

Figure 2a-b. An example of the probability distribution of possible upcoming word choices (a) when recipients were given (*Chelswu-ka kyengchal-eykey* meaning Chelswu-$_{NOM}$ policeman-$_{DAT}$), and (b) when themes/patients were given (*Chelswu-ka sinpwunchun-ul* meaning Chelswu-$_{NOM}$ ID-$_{ACC}$). Each piece of the pie represents one possible word choice with bigger size of the piece being higher probability.

In sum, as we hypothesized, the predictability patterns for the upcoming likely information were not different between older readers and young readers in terms of both structures and words. No aging effect suggests that older readers developed upcoming information, as good as young readers, by employing their linguistic and world knowledge. However, we also observed a significant aging effect, especially in the predictive patterns of unlikely choices in terms of structures and words. Older readers tend to have stronger predictability to unlikely structural choices than young readers, whereas older readers tend to have weak predictability to unlikely word choices.

7. General Discussion

The purpose of the present study was to investigate whether older readers would develop upcoming information, as good as young readers, by employing their linguistic and world knowledge. For this purpose, we compared the responses from older and young readers obtained from two offline tasks (cloze task, listing task) in which there used two kinds of sentence fragments (when agents and recipients were given or when agents and patients were given). Two types of predictabilities (structural predictability, lexical predictability) were computed by using two types of computation measurements (cloze probability, uncertainty).

Our hypothesis was supported. First, there was no aging effect in producing likely structural choices, meaning that the probability likely targets generated by younger and older readers show considerable similarities. Like young readers, older readers produced theme NPs as most-likely structures when agents and recipients were given (Figure 1a) and verbs as most-likely structures when agents and themes were given (Figure 1b). Second, there was no aging effect either in producing most-likely word choices for given contexts, meaning that the most predictable words to older readers were also the most predictable to young readers. In the total of 36 listing task sentences, the most-likely (i.e., the first choice) word generated by younger and older readers was the same at 58%, and the similarity corresponding to the not-most-likely but slightly less-likely words (i.e., the second and third choice) increased up to 75%. These results indicated that there was a considerable overlap of semantic expectation for given contexts across the two age groups.

No aging effect in generating the predictive representation of likely upcoming choice support our tentative speculation: The difficulty observed in older readers' predictive sentence processing is fundamentally dependent on their declined working memory, which in turn leads to more difficulty of operating the combinatorial system (System 2) but less difficulty of operating the fast and automatic system (System 1). Therefore, when using their semantic knowledge system (associated with System 1) in detecting the semantic properties of upcoming information, the knowledge and linguistic competence of older readers should be as good as those of young readers. Furthermore, the results that we observed are in line with the existing claims that older readers successfully employ their linguistic and world knowledge in the processing of previous context. That being said, older readers' use of knowledge and linguistic experiences is intact in language processing (e.g., Payne et al., 2012; 2014). Of course, the use of predictive representation during online sentence processing (associated with System 2) might be delayed (e.g., Stine and Wingfield, 1994; Lash, et al. 2013; Madden, 1988; Stine-Morrow et al., 2008), but this is beyond the scope of the current study but that of our future study.

We also observed a significant aging effect, especially regarding the predictive patterns of unlikely choices, in both structural predictability and lexical (contextual) predictability. First, we found the significant difference of structural uncertainty between the age groups which was mainly attributed to the difference of probability distribution of unlikely structural choices. For instance, as shown in Figure 1a, the cloze probabilities of unlikely structural choices, generated by young readers, were gradually lowered step by step, but this was not found in the probability patterns of older readers. This slight difference indicates that there would a less competition among unlikely choices generated by young readers, resulting in more certain structural expectation, especially for uncertain choices. Moreover, the aging effect was also elicited in the predictability of target structural choice when it was extremely unlikely (i.e., the predictability of recipient NPs as an upcoming choice when agents and themes are given). As shown in Figure 1b, recipients' NPs showed a greater tolerance for older readers than younger readers. However, we are careful in assigning too much significance on the aging effect found in the prediction of unlikely choices, in that the overall structural uncertainty probability of the two age groups was very low and standard deviations were large as well. Rather, we put a lot more emphasis on no aging effect found in the prediction of the likely structural choices, suggesting that older readers' linguistic knowledge is well used for the prediction of upcoming structural information as successfully as young readers' knowledge.

Of interest, the opposite pattern emerged in the lexical predictability of unlikely words for given contexts. Young readers produced a wider range of possible word choices, making the given context less constraining and less predictable (Figure 2a). In contrast, older readers produces a less wide range of possible word choices, making the given context more constraining and more predictable (also Figure 2a). We note that the older readers showed a much clearer picture of the first, second, and third word choice compared to the younger readers. In other words, the diversity of words generated in the context of a sentence was more certain than that of younger readers.

In short, older readers seemed to be more ready to encounter unlikely structures as long as they were not sure which structure would like to appear (c.f., recall how our cloze task was conducted), but they tended to constrain their semantic expectation to words when they knew which structure appear (c.f., recall how our listing task was conducted). These age differences are consistent with previous studies that show older adults use a conservative strategy in lexical-decision tasks compared to younger adults (Ratcliff et al., 2004). There is a need to investigate in depth the younger and older readers' differences in strategy by examining the use of contextual information using on-line sentence processing.

The limitations of the present study are as follows. First, the generalization of the study results is limited owing to the wide age range and small sample size of the older group. Therefore, future studies need to verify the results of the present study by including a larger number of older adults. Second, the results indicated that intra-individual performance-variability increases with advancing age. It would therefore be necessary to compare predictive representation for upcoming linguistic input within older adults by splitting older adults into young-old (60-75 years) and old-old (75+ years) groups.

In sum, the results of the present study revealed that younger and older readers exhibit considerable similarities in overall structural and lexical predictability in Korean sentence comprehension. Older readers' use of contextual information in predicting sentence structures and words measured by using off-line measure is similar to that of younger readers. We speculate that older readers might have higher sensitivity of the contextual information which leads to reader's being much denser consolidation to context and eventually their stronger commitment for upcoming information. This might be operated by the fast and automatic system that older readers have long been operating based on their knowledge and experiences. How this knowledge-based system will be implemented in the processing of combinatorial information is our future work. Under the circumstances, where there is a paucity of research related to aging in Korean language processing, this study holds significance in that it found similar patterns of context-based semantic knowledge representation for upcoming linguistic input in Korean older and younger adults.

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Received on: June 21, 2018
Revised on: July 23, 2018
Accepted on: July, 26, 2018